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Dear Toyota Material Handling North America Hiring Team,

I am excited to apply for the Firmware Development Engineering Internship Toyota Material Handling North America. I am currently pursuing a B.S. in Computer Science in Real-Time Interactive Simulation at DigiPen Institute of Technology, where I have developed a strong foundation in C++, C, systems programming, computer graphics, and software architecture. I am particularly interested in this position because it combines software development with low-level system behavior, testing, debugging, and hardware-oriented problem solving.

As the Producer and Engine/Graphics Engineer for *Dragon Tactics*, a 2D turn-based tactical RPG developed by a five-person team, I worked extensively with a custom C++ game engine. I refactored the engine to remove its previous rendering dependency and integrate OpenGL directly, developed core rendering systems, and helped design communication between engine and gameplay systems using an EventBus. This work required careful debugging, understanding how multiple subsystems interacted, and building reliable software components that other parts of the project could depend on.

In a personal Game Boy Advance project, I explored lower-level hardware-oriented programming by working directly with the GBA's graphics and audio subsystems. Building on concepts I had learned in computer graphics, I implemented texture rendering through the GBA's PPU and worked directly with its sound hardware to produce audio. This project strengthened my interest in understanding how software interacts with underlying hardware and gave me practical experience debugging behavior close to the system level.

I would be excited to apply my C/C++ background, debugging experience, and systems-oriented mindset to firmware testing, utility development, and embedded software development at Toyota Material Handling. I am especially interested in learning more about the full firmware development process and contributing to tools and tests that improve product reliability.

Thank you for considering my application. I would welcome the opportunity to contribute my C/C++ and software engineering experience to Toyota Material Handling North America while continuing to develop my skills in firmware and embedded systems engineering.

Sincerely,

Taekyung (Jean) Ho